

MOSSTRAND II, Norway

WMO: 014500

Lat: 59.839N

Lon: 8.179E

Elev: 977

StdP: 90.13

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.9	-17.6	-23.2	0.5	-19.1	-20.9	0.7	-17.2	19.1	-1.8	17.0	-2.4	1.1	320	0.800

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	21.4	12.9	19.4	12.0	17.3	11.2	13.9	19.2	13.0	17.4	12.0	15.9	3.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.1	9.9	14.8	11.1	9.2	14.2	10.1	8.7	13.3	42.1	19.5	39.3	17.3	36.9	16.0	17.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	14.7	12.3	10.4	DB	-22.5	24.0	2.5	1.9	-24.2	25.3	-25.7	26.4	-27.1	27.5	-28.9	28.8
(5)				WB	-22.8	15.2	2.4	1.2	-24.6	16.1	-26.0	16.8	-27.3	17.5	-29.0	18.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.8	-7.0	-6.4	-3.4	0.1	5.0	9.3	11.8	10.2	7.2	2.1	-2.4	-5.1
	DBStd	7.57	4.92	4.99	4.45	3.61	4.09	2.93	3.13	2.18	2.68	3.06	3.92	5.07
	HDD10.0	3126	527	459	416	298	164	47	14	23	91	246	371	469
	HDD18.3	6022	786	692	674	548	413	270	202	251	334	503	621	728
	CDD10.0	145	0	0	0	0	9	27	71	31	7	1	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	13	0	0	0	0	0	1	12	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.0	4.4	4.2	4.5	3.6	3.3	3.8	3.5	3.6	4.2	4.1	4.2	4.5
Precipitation	PrecAvg	790	75	47	42	38	51	68	89	92	80	77	69	63
	PrecMax	1085	164	127	112	82	138	169	185	164	151	199	178	147
	PrecMin	574	11	2	3	11	2	10	27	8	25	12	17	5
	PrecStd	121	39	30	25	22	31	37	37	38	31	44	38	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	5.1	9.1	11.9	20.4	22.2	25.1	20.3	18.0	12.3	7.1	5.2
		MCWB	2.0	2.2	4.1	5.8	11.5	12.2	13.8	13.0	11.6	8.5	4.8	2.9
	2%	DB	2.3	3.2	6.0	9.4	17.8	20.2	22.6	18.0	15.3	10.1	5.1	3.9
		MCWB	0.9	0.9	2.5	4.5	10.7	11.0	13.7	12.2	10.3	6.7	3.6	2.3
	5%	DB	0.9	1.8	3.9	7.1	14.7	17.9	20.6	16.0	12.9	8.0	3.8	2.7
		MCWB	-0.3	0.1	1.3	3.2	8.6	10.1	13.1	11.1	9.3	5.7	2.6	1.4
	10%	DB	-0.6	0.3	2.4	5.4	12.0	15.5	18.1	14.3	11.4	6.5	2.6	1.0
		MCWB	-1.4	-1.1	0.2	2.5	7.0	9.1	12.2	10.4	8.6	4.6	1.5	0.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	2.9	4.6	6.9	12.7	13.2	15.9	14.0	12.7	10.6	5.4	4.3
		MCDB	3.3	4.5	8.3	10.9	18.8	19.8	21.1	17.9	15.9	12.2	6.6	4.9
	2%	WB	1.0	1.4	2.8	5.1	10.9	12.0	14.6	12.9	11.0	7.4	4.0	2.6
		MCDB	2.0	2.8	5.3	8.2	16.4	17.5	20.2	16.1	13.8	8.8	5.1	3.5
	5%	WB	-0.1	0.2	1.5	3.7	9.3	11.1	13.7	12.2	10.0	6.2	2.8	1.3
		MCDB	0.6	1.4	3.6	6.7	13.7	16.0	19.0	15.3	12.3	7.7	3.6	2.3
	10%	WB	-1.4	-1.1	0.3	2.7	7.8	10.0	12.7	11.1	9.0	4.8	1.5	-0.1
		MCDB	-0.6	0.3	2.2	5.2	11.3	14.6	17.3	13.6	11.2	6.2	2.5	0.8
Mean Daily Temperature Range	5% DB	MDBR	5.4	6.3	7.4	8.2	8.8	9.2	8.5	7.4	6.6	5.6	4.6	5.2
		MCDBR	6.1	7.1	8.3	10.5	13.1	13.6	13.0	10.0	9.5	8.5	5.6	5.4
		MCWBR	5.3	5.8	5.5	6.3	7.1	6.4	5.5	5.1	5.5	5.7	4.7	5.0
	5% WB	MCDBR	5.6	6.1	7.6	9.3	11.8	11.2	10.8	8.5	8.4	6.5	5.1	5.4
		MCWBR	5.1	5.4	5.2	5.7	6.7	5.9	5.1	5.0	5.2	5.3	4.5	5.2
Clear-Sky Solar Irradiance	taub	0.208	0.244	0.272	0.307	0.312	0.307	0.322	0.316	0.289	0.273	0.223	0.181	
	taud	2.202	2.300	2.285	2.300	2.460	2.532	2.495	2.501	2.571	2.515	2.275	2.078	
	Ebn at Noon	630	795	872	897	919	928	903	884	859	760	600	508	
	Edh at Noon	38	63	89	106	98	93	94	87	68	52	36	28	
All-Sky Solar Radiation	RadAvg	0.32	0.96	2.41	4.04	4.79	5.28	4.70	3.71	2.41	1.14	0.42	0.21	
	RadStd	0.07	0.22	0.29	0.62	0.66	0.51	0.61	0.30	0.31	0.21	0.10	0.02	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page